



Report on

RELIABILITY PREDICTION

OF

Camera Model No.: HC-IPC-TI4312DVZK-N8-O
(CCTV CAMERA)

as per Telcordia SR-332, Issue 4



CENTRE FOR RELIABILITY,
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1.0 SCOPE

- 1.1** Service request no. : 2602311
- 1.2** Name & Address of the Client : Hi-Focus Electronics India Pvt Ltd
3, Sunguvar Agraharam St,
Chindatripet, Chintadripet, Chennai,
Tamil Nadu 600002.
- 1.3** Description : **CCTV CAMERA**
Make : Hi-Focus Electronics India Pvt Ltd
Model No. : HC-IPC-TI4312DVZK-N8-O
Serial No. : HFP25100543120730
- 1.4** Project : Received Date : 11-05-2026
Completed Date : 09-06-2026
- 1.5** (a) Standard/ Analysis / Test Procedure Ref. : Telcordia SR-332, Issue 4
(b) Scope : Reliability Prediction by Parts Count Method

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2.0 INTRODUCTION

This report presents the results of Reliability Prediction Analysis performed on **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** as per Telcordia SR-332, Issue 4 as depicted in Figure 1.0, 2.0, 3.0, 4.0, 5.0, 6.0. and 7.0 This exercise was performed under **Ground, Fixed, Controlled (GB)** environment as requested by **Hi-Focus Electronics India Pvt Ltd.**

Design Reliability Prediction is the process of quantitatively assessing whether a proposed or actual equipment design will meet a specified reliability requirement. Design Reliability Prediction has various applications in the design and development process including substantiating quantitative design comparisons with competitive products, selecting warranty periods, advertising and marketing. It provides guidance, relative to expected inherent reliability of a given design.



Figure 1.0 (a & b) Front & Top view of camera

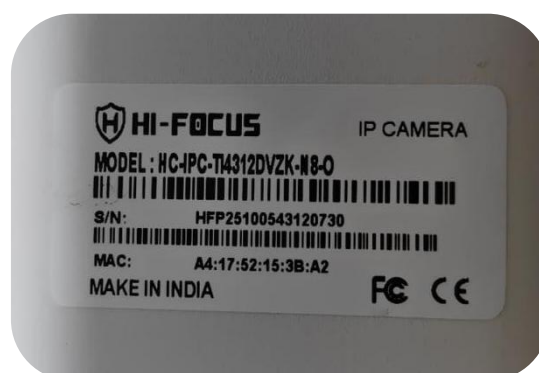


Figure 2.0 (a) Markings on camera

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Camera Model No.: HC-IPC-TI4312DVZK-N8-O**

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09-06-2026**



Figure 3.0 (a & b) Front & Rear View of PCB 1



Figure 4.0 (a & b) Front & Rear View of PCB 2

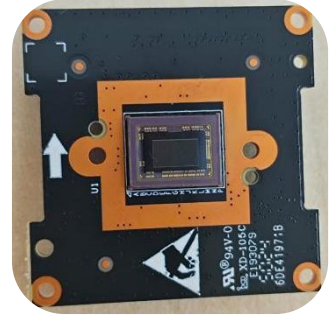


Figure 5.0 (a & b) Front & Rear View of PCB 2



Figure 6.0 (a & b) Motorized Lens



Figure 7.0 (a) IR Board & LEDs

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3.0 APPLICABLE DOCUMENTS AND TOOLS

Telcordia SR-332, Issue 4 - Reliability Prediction Procedure for Electronic Equipment

4.0 RELIABILITY PREDICTION OBJECTIVES

Objectives of this analysis are to estimate the Mean Time between Failure (MTBF) of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** using Telcordia SR-332, Issue 4 and to obtain quantitative design information as applicable. The main objectives are:

- a. Provide an estimate of the steady-state failure rates for device design based on generic reliability parameters and obtain feasibility evaluation.
- b. Provide feedback for designers.
- c. Identify potential problem areas during design review.
- d. Provide MTBF of the system and sub-assemblies and failure rate of system/sub-assemblies/ Components.

5.0 SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method - ASSUMPTIONS

Reliability Prediction using SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method for reliability prediction operates under the following assumptions: assumes a serial relationship between parts of the system, in which, failure of any part constitutes system failure. The model also assumes that

- a. At the device level, it is assumed that no detailed reliability data is available. Predictions are based on generic reliability parameters.
- b. Predictions are made using generic reliability parameters as specified in the relevant sections of SR-332, which are applied to estimate steady state failure rates.
- c. At the unit level, failure rate predictions are based solely on the failure rates of the individual devices that compose the unit, without considering interactions between devices.
- d. This method primarily estimates steady-state failure rates and does not specifically account for infant mortality failures (early-life failures) or wear out failures (end-of-life failures). It assumes a constant failure rate.

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6.0 SR-332, Method I: Unit Steady-State Failure Rate Parts Count Method: Factors

Method I: Unit Steady-State Failure Rate Using the Parts Count Method primarily estimates reliability by calculating the steady-state failure rate of a device using a set of generic parameters. It takes into account the following factors:

- 6.0.1 Number of component of each Type
- 6.0.2 Mean Generic Steady-State Failure Rate
- 6.0.3 Temperature factor
- 6.0.4 Environmental factor
- 6.0.5 Quality factor
- 6.0.6 Electrical Stress Factor

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7.0 ENVIRONMENTAL CONDITIONS

7.1 STORAGE AND OPERATIONAL ENVIRONMENT

The Reliability Prediction has been performed using **Ground, Fixed, Controlled (GB)** Environment based on the operational conditions of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O**.

GROUND, FIXED, CONTROLLED (GB):

The environment represents a stationary, land-based installation located in a protected or climate-controlled facility. It is characterized by low vibration and shock stresses, low atmospheric variations, and low temperature-cycling stresses, with the understanding that a building or ruggedized enclosure provides consistent protection from weather and contaminants.

TEMPERATURE

Reliability Prediction has been performed on **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** at the **Ambient Temperature of 40 °C for Ground, Fixed, Controlled (GB) Environment**.

7.2 QUALITY LEVEL : Level- 1

The unit has been evaluated under the commercial-grade component Quality level-1 as mentioned by the client in bill of material. The components used in the unit were procured without undergoing thorough device qualification or lot-to-lot controls by the equipment manufacturer.

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8.0 REPORTS

Report includes the following information-

- a) Components/ Parts descriptions
- b) Failure Rate
- c) Pi (π) Factors
- d) Parameters details

Refer Annexure- I.

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9.0 System Analysis Summary

The overall estimated MTBF and Failure Rate as per Telcordia SR-332 of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O**, submitted for reliability prediction analysis by **Hi-Focus Electronics India Pvt Ltd** are summarized as follows:

SYSTEM: Camera Model No.: **HC-IPC-TI4312DVZK-N8-O**

Environment	Ambient Temperature	Failure Rate (Failures/Million Hours)	MTBF (Hrs)
Ground, Fixed, Controlled (GB)	40°C	4.117048	242892.49

- Design reliability prediction analysis was carried out based on the failure rates of components used in the camera (refer Annexure I), following a physical verification of components and in accordance with Telcordia SR-332, Issue 4.
- The camera contains three (03) nos of synchronous/stepper/shutter motors. However, failure rate data for these motors is not available in the Telcordia standard or in the reliability prediction software used. Therefore, for the purpose of the design reliability prediction, the failure rate for each motor has been conservatively assumed as one (1) FIT.
- MTBF (Mean Time Between Failures) calculations have been performed using the component details—such as type, quantity, electrical and environmental parameters, stress levels, and quality grades—as provided by the client in their submitted Bill of Materials.

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10.0 Abbreviation Used:

- a) MTBF : Mean Time Between Failure
- b) λG : Base failure rate for the generic part (Ground fixed base rate)
- c) σG : Standard deviation or uncertainty in base failure rate (generic)
- d) πQ : Quality factor (related to part/process quality level)
- e) πE : Environmental factor (based on the operating environment, e.g., ground benign, airborne)
- f) πT : Temperature factor (influenced by ambient/operating temperature)
- g) πS : Stress factor (e.g., voltage or power stress ratio)
- h) λSS : Subsystem or system-level failure rate (or stress-adjusted failure rate)
- i) σSS : Standard deviation or uncertainty in system/subsystem failure rate

- End of Report -

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Annexure - I

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Reliability Prediction Report

Name	CCTV CAMERA	Calculation Model	Telcordia Issue 4
Description	Make : HI-FOCUS	Environment	Ground, Fixed, Controlled (GB)
	Model Number : HC-IPC-TI4312DVZK-N8-O	Ambient Temperature, Celsius	40.00
	Serial Number : HFP25100543120730	Operating Profile	Telecommunications
	Job No : 10TE202601266	Failure Rate Units	Failures/Million Hours
Quantity	1	Failure Rate	4.117048
Reference Designator		MTBF Units	Hours
		MTBF	242892.49

PARTS

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.1	R0402RXX472 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5,R15,R28,R36, R65,R72,R90,R91	4.7K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.2	R0402RXX330 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R20,R21,R42, R43,R57,R58	33 Ω $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.3	R0402RXX202 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R74,R94	2K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.4	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R24	22K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.5	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R70,R95	1K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.6	R0402RXX105 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R27	1M $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: mohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.7	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R7,R11,R29,R30, R39,R93	10K $\pm$ 1%-0402-(-55°C~155°C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.8	R0402RXX104 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R31,R89,R96	100K±1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.9	R0402RXX241 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R38	240Ω±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.10	R0402JXX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R32,R34,R44	0Ω±5%-0402-(-55°C~155°C)	3	0.000152	4.562529E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.53301804, λG: 0.08, σG: 0.02, λSS: 0.12792433, σSS: 0.031981083, Temperature: 40
1.11	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R33,R37,R69,R88	18K±1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.12	R0402RXX152 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R92	1.5K $\pm$ 1%-0402- (-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1.5, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.13	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R18	3.3K $\pm$ 1%-0402- (-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.14	R0402RXX473 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R87	47K $\pm$ 1%-0402- (-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.15	R0402RXX751 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R35	750 Ω $\pm$ 1%-0402- (-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.16	R0402RXX153 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R73	15K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.17	R0402RXX471 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R64,R100,R706	470Ω±1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.18	R0402RXX474 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5	470K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.19	R0402RXX472 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R42,R67,R151,R174,R274,R277,R431,R512,R513,R598,R703,R712,R737,R742,R746,R787,R788,R789,R790,R791,R792,R803,R809,R815,R818,R822,R823,R824,R859,R860,R871,R872,R873,R874,R886,R897,R1039,R1040	4.7K±1%-0402-(-55°C~155°C)	38	0.000162	0.006167	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.20	R0402RXX330 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R36,R75,R76,R77,R78,R87,R89,R90,R91,R105,R181,R184,R187,R202,R230,R231,R234,R514,R515,R804,R811,R850,R853,R854,R855,R856,R857,R858,R864,R865,R866,R867,R868	33 Ω ±1%-0402-(-55°C~155°C)	33	0.000163	0.005374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.21	R0402RXX202 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R45,R63,R168,R170,R173,R186,R243,R276,R299,R315,R728,R896	2K±1%-0402-(-55°C~155°C)	12	0.000169	0.002022	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.22	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R724	22K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.23	R0402RXX222 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R61,R94,R563,R730,R915	2.2K±1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.24	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R69,R345,R734, R744,R748,R793 ,R830,R1003,R1 005	1K±1%-0402-(- 55°C~155°C)	9	0.000171	0.001537	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.25	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R14,R39,R44,R7 3,R709,R722,R7 36,R738,R740,R 741,R794,R831, R1013,R1015,R1 016	10K±1%-0402-(- 55°C~155°C)	15	0.000167	0.002505	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.26	R0402RXX104 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R721,R801	100K±1%-0402- (-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.27	R0402RXX241 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1007,R1008	240Ω±1%-0402- (-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.28	CR-05JL7---10R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R262,R899	10Ω±5%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, λG: 0.08, σG: 0.02, λSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.29	CR-05FL7--100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R242	100Ω±1%-0805-(-55°C~155°C)	1	0.000185	1.850080E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, λG: 0.08, σG: 0.02, λSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.30	CR-05JL7---OR	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R23	0Ω±5%-0805-(-55°C~155°C)	1	0.000168	1.684770E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.52750338, λG: 0.08, σG: 0.02, λSS: 0.12660081, σSS: 0.031650203, Temperature: 40
1.31	R0402JXX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1,R2,R7,R9,R11,R15,R16,R17,R59,R66,R70,R82,R86,R88,R166,R546,R723,R727,R735,R739,R750,R819,R833,R912,R1001,R1002	0Ω±5%-0402-(-55°C~155°C)	26	0.000136	0.003535	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.53301804, λG: 0.08, σG: 0.02, λSS: 0.12792433, σSS: 0.031981083, Temperature: 40
1.32	R0402RXX100 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R112	10Ω±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.33	R0402RXX101 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R4,R52,R101, R106,R814,R817	100Ω±1%-0402- (-55°C~155°C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.34	CR-03FL7--- 75R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R780,R783	75Ω±1%-0603-(- 55°C~155°C)	2	0.000566	0.001133	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 75, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 1.9155408, λG: 0.08, σG: 0.02, λSS: 0.4597298, σSS: 0.11493245, Temperature: 40
1.35	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R729	18K±1%-0402-(- 55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.36	R0402RXX333 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R743,R745	33K±1%-0402-(- 55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.37	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R717,R832,R898	3.3K±1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.38	R0402RXX751 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R726	750Ω±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.39	R0402RXX153 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R704,R720,R725 ,R732,R747	15K±1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.40	R0402RWX2R 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R10,R12,R13	2.2Ω±1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.41	RS-05FL7--- OR2	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R80,R585	200mR±1%-0805-(-55°C~155°C)	2	0.000260	5.193472E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.2, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.87809543, λG: 0.08, σG: 0.02, λSS: 0.2107429, σSS: 0.052685726, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.42	CR0402FF56R0G	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1014,R1017	56R $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 56, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.43	CR0402FF680OG	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1018,R1019,R1020,R1021	680R $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 680, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.44	RL0805FR-070R3L	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R79,R584	300mR $\pm$ 1%-0805-(-55°C~155°C)	2	0.000260	5.193472E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.3, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.87809543, λ G: 0.08, σ G: 0.02, λ SS: 0.2107429, σ SS: 0.052685726, Temperature: 40
1.45	R0402RXX510XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R35	51Q $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 51, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.46	R0402RXX474XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R37	470K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

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Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.47	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R26,R36	22K±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.48	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R29	1K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.49	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R13,R22	10K±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.50	CR-05FL7--- 10K	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R9,R10	10K±1%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.51	CR-05FL7--100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R2,R3	100Ω±1%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, λG: 0.08, σG: 0.02, λSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.52	CR-05JL7---OR	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R17	0Ω±5%-0805-(-55°C~155°C)	1	0.000168	1.684770E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.52750338, λG: 0.08, σG: 0.02, λSS: 0.12660081, σSS: 0.031650203, Temperature: 40
1.53	RC0805JR-071RL	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R30,R32,R33,R34	1Ω±5%-0805-(-55°C~155°C)	4	0.000162	6.468270E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, λG: 0.08, σG: 0.02, λSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.54	R0402JXX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R14	0Ω±5%-0402-(-55°C~155°C)	2	0.000158	3.152521E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.53301804, λG: 0.08, σG: 0.02, λSS: 0.12792433, σSS: 0.031981083, Temperature: 40
1.55	R0402RXX100 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R21	10Ω±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.56	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R20	18K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.57	R0402RXX333 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R15	33K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.58	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R6,R7,R11,R12, R27	3.3K $\pm$ 1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.59	R0402RXX473 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R19,R25	47K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.60	R0402RXX201XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R28	200 Ω ±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 200, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.61	CC0402JRNPO9BN100	Capacitor	Ceramic	General	C81	10pF±5%-50V-NPO-0402	1	0.000055	5.526646E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40
1.62	CC0402KRX7R9BB102	Capacitor	Ceramic	General	C47	1nF±10%-50V-X7R-0402	1	0.000048	4.815829E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.63	0402X105K6R3CT	Capacitor	Ceramic	General	C30,C31,C32,C58	1uF±10%-6.3V-X5R-0402	4	0.000090	3.589336E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.28109921, λ G: 0.1, σ G: 0.01, λ SS: 0.084329762, σ SS: 0.0084329762, Temperature: 40
1.64	CC0402KRX7R7BB104	Capacitor	Ceramic	General	C14,C33,C34,C38,C54,C56,C59,C60,C61,C62,C63,C64,C70,C80,C88,C89,C93,C99,C457	100nF±10%-16V-X7R-0402	19	0.000093	0.001760	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.65	0402B103K50OCT	Capacitor	Ceramic	General	C11,C13,C20,C44	10nF±10%-50V-X7R-0402	4	0.000045	1.813783E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.66	0402X225M6 R3CT	Capacitor	Ceramic	General	C12	2.2uF±20%-6.3V-X5R-0402	1	0.000288	2.876279E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.67	CL05A475MQ 5NRNC	Capacitor	Ceramic	General	C3,C6,C7,C9,C15,C17,C19,C21,C23,C25,C27,C36	4.7uF±20%-6.3V-X5R-0402	12	0.000264	0.003173	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.68	0402X224K10 ONT	Capacitor	Ceramic	General	C86,C104	220nF±20%-10V-X5R-0402	2	0.000163	3.259277E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.4980773, λ G: 0.1, σ G: 0.01, λ SS: 0.14942319, σ SS: 0.014942319, Temperature: 40
1.69	CL10A106KP8 NNNC	Capacitor	Ceramic	General	C39,C57,C71,C436	10uF±10%-10V-X5R-0603	4	0.000319	0.001277	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: 1, λ G: 0.1, σ G: 0.01, λ SS: 0.3, σ SS: 0.03, Temperature: 40
1.70	CL10A226MQ 8NRNC	Capacitor	Ceramic	General	C37,C40,C42,C51,C55,C79,C90,C92,C94,C96,C103,C105,C107,C232,C233	22uF±20%-6.3V-X5R-0603	15	0.000342	0.005126	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	π Q: 3, π E: 1, π T: 1, π S: 1.1025427, λ G: 0.1, σ G: 0.01, λ SS: 0.33076281, σ SS: 0.033076281, Temperature: 40
1.71	CC0402JRNPO 9BN100	Capacitor	Ceramic	General	C723,C825	10pF±5%-50V-NPO-0402	2	0.000053	1.068553E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.72	CC0402JRNPO9BN180	Capacitor	Ceramic	General	C815,C822,C824	18pF±5%-50V-NP0-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.73	0402N270J50OCT	Capacitor	Ceramic	General	C23	27pF-50V-NP0-±5%-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.74	0402N330J50OCT	Capacitor	Ceramic	General	C12,C18,C31,C64	33pF-50V-NP0-±5%-0402	4	0.000054	2.154621E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.75	CC0402JRNPO9BN101	Capacitor	Ceramic	General	C17	100pF±5%-50V-NP0-0402	1	0.000066	6.576536E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.19398004, λ G: 0.1, σ G: 0.01, λ SS: 0.058194013, σ SS: 0.0058194013, Temperature: 40
1.76	CC0402KRX7R9BB102	Capacitor	Ceramic	General	C8,C11,C32,C33,C52,C76	1nF±10%-50V-X7R-0402	6	0.000045	2.690611E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.77	0402B332K50OCT	Capacitor	Ceramic	General	C93,C489	3.3nF±10%-50V-X7R-0402	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.78	CC1206KKX7RDBB102	Capacitor	Ceramic	General	C6,C260,C301,C550	1nF±10%-2KV-X7R-1206	4	0.000042	1.660743E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.13006122, λ G: 0.1, σ G: 0.01, λ SS: 0.039018367, σ SS: 0.0039018367, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.79	0402N221J50 OCT	Capacitor	Ceramic	General	C705,C716,C731	220pF±5%-50V-NPO-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.80	CC0805KKX5R7BB106	Capacitor	Ceramic	General	C15,C77,C277,C331	10uF±10%-16V-X5R-0805	4	0.000096	3.829139E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.81	0603N821J50 OCT	Capacitor	Ceramic	General	C381,C382	820pF±5%-50V-NPO-0603	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.82	0402X105K6R3CT	Capacitor	Ceramic	General	C739,C801,C821	1uF±10%-6.3V-X5R-0402	3	0.000091	2.717081E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.28109921, λ G: 0.1, σ G: 0.01, λ SS: 0.084329762, σ SS: 0.0084329762, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.83	CC0402KRX7R7BB104	Capacitor	Ceramic	General	C5,C9,C10,C19,C28,C40,C44,C57,C66,C70,C86,C119,C137,C253,C254,C276,C282,C334,C711,C715,C719,C726,C733,C738,C741,C747,C751,C753,C802,C804,C806,C810,C811,C813,C814,C817,C818,C819,C820,C823,C827,C829,C830,C831,C832,C835,C836,C850,C852,C853,C854,C855,C858,C859,C860,C998,C1002,C1003,C1004,C1006,C1012,C1013,C1014,C1015,C1016,C1017,C1018,C1019,C1020,C1021,C1022,C1023,C1024,C1025,C1026,C1028,C1029,C1030,C1031,C1032,C1033,C1034,C1035,C1036,C1037,C1038,C1039,C1040,C1041,C1102	100nF±10%-16V-X7R-0402	90	0.000091	0.008206	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.29987937, λG: 0.1, σG: 0.01, λSS: 0.08996381, σSS: 0.008996381, Temperature: 40
1.84	0402B103K50OCT	Capacitor	Ceramic	General	C56,C1042	10nF±10%-50V-X7R-0402	2	0.000046	9.295127E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.14204662, λG: 0.1, σG: 0.01, λSS: 0.042613985, σSS: 0.0042613985, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.85	0402X225M6R3CT	Capacitor	Ceramic	General	C24,C98,C816	2.2uF±20%-6.3V-X5R-0402	3	0.000274	8.214552E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.84984745, λG: 0.1, σG: 0.01, λSS: 0.25495424, σSS: 0.025495424, Temperature: 40
1.86	0805X226M100CT	Capacitor	Ceramic	General	C69	22uF±20%-10V-0805	1	0.000338	3.384465E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	πQ: 3, πE: 1, πT: 1, πS: 1, λG: 0.1, σG: 0.01, λSS: 0.3, σSS: 0.03, Temperature: 40
1.87	UT1E101M0607VG	Capacitor	Electrolytic	Aluminum	CE1	100uF±20%-25V-105°C-Φ6.3*7.7mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 100, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.95313379, λG: 0.73, σG: 0.17, λSS: 2.087363, σSS: 0.48609823, Temperature: 40
1.88	CL05A475MQ5NRNC	Capacitor	Ceramic	General	C73,C75,C186,C805,C807,C808,C809,C828,C857	4.7uF±20%-6.3V-X5R-0402	9	0.000266	0.002393	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.84984745, λG: 0.1, σG: 0.01, λSS: 0.25495424, σSS: 0.025495424, Temperature: 40
1.89	CL10A106KP8NNNC	Capacitor	Ceramic	General	C1,C16,C41,C296,C704,C714,C721,C728,C735,C746,C812,C826,C851,C856,C1001,C1027	10uF±10%-10V-X5R-0603	16	0.000310	0.004954	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	πQ: 3, πE: 1, πT: 1, πS: 1, λG: 0.1, σG: 0.01, λSS: 0.3, σSS: 0.03, Temperature: 40
1.90	CL21A106KAYNNNE	Capacitor	Ceramic	General	C289,C333,C743	10uF±10%-25V-X5R-0805	3	0.000297	8.904935E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.92127196, λG: 0.1, σG: 0.01, λSS: 0.27638159, σSS: 0.027638159, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.91	CL10A226MQ8NRNC	Capacitor	Ceramic	General	C20,C21,C197,C706,C710,C717,C730,C732,C737,C750,C803	22uF±20%-6.3V-X5R-0603	11	0.000344	0.003779	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	πQ: 3, πE: 1, πT: 1, πS: 1.1025427, λG: 0.1, σG: 0.01, λSS: 0.33076281, σSS: 0.033076281, Temperature: 40
1.92	0402N270J50OCT	Capacitor	Ceramic	General	C8	27pF-50V-NP0-±5%-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16873936, λG: 0.1, σG: 0.01, λSS: 0.050621808, σSS: 0.0050621808, Temperature: 40
1.93	CC1206KKX7RDBB102	Capacitor	Ceramic	General	C7,C9,C10	1nF±10%-2KV-X7R-1206	3	0.000042	1.257161E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.13006122, λG: 0.1, σG: 0.01, λSS: 0.039018367, σSS: 0.0039018367, Temperature: 40
1.94	0402N221J50OCT	Capacitor	Ceramic	General	C17,C19	220pF±5%-50V-NP0-0402	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16873936, λG: 0.1, σG: 0.01, λSS: 0.050621808, σSS: 0.0050621808, Temperature: 40
1.95	1206B104K201NT	Capacitor	Ceramic	General	C2,C3,C12	100nF±10%-200V/250V-X7R-1206	3	0.000111	3.328779E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.3443833, λG: 0.1, σG: 0.01, λSS: 0.10331499, σSS: 0.010331499, Temperature: 40
1.96	1206B105K101CT	Capacitor	Ceramic	General	C5	1uF±10%-100V-X7R-1206	1	0.000312	3.123403E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.92127196, λG: 0.1, σG: 0.01, λSS: 0.27638159, σSS: 0.027638159, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.97	VZH221M1ETR-0810	Capacitor	Electrolytic	Aluminum	CE2	220uF±20%-25V-105°C-Φ8*10mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 220, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.95313379, λG: 0.73, σG: 0.17, λSS: 2.087363, σSS: 0.48609823, Temperature: 40
1.98	CC0402KRX7R7BB104	Capacitor	Ceramic	General	C11	100nF±10%-16V-X7R-0402	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.29987937, λG: 0.1, σG: 0.01, λSS: 0.08996381, σSS: 0.008996381, Temperature: 40
1.99	UL1J470M081OVG	Capacitor	Electrolytic	Aluminum	CE1	47uF±20%-63V-105°C-Φ8*10mm	1	0.005369	0.005369	Type: Aluminum, Quality Level: Level I, Capacitance: 47, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	Level I	πQ: 3, πE: 1, πT: 1, πS: 1.8749302, λG: 0.73, σG: 0.17, λSS: 4.1060971, σSS: 0.9562144, Temperature: 40
1.100	CL21A106KAYNNNE	Capacitor	Ceramic	General	C6,C14,C15	10uF±10%-25V-X5R-0805	3	0.000297	8.904935E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.92127196, λG: 0.1, σG: 0.01, λSS: 0.27638159, σSS: 0.027638159, Temperature: 40
1.101	TCC0805X7R102K201DT	Capacitor	Ceramic	General	C21	1nF±20%-200V-X7R-0805	1	0.000056	5.572151E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 200, Percentage: 6	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 200, Percentage: 6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16463901, λG: 0.1, σG: 0.01, λSS: 0.049391704, σSS: 0.0049391704, Temperature: 40
1.102	11-1832002-330	Connection	Multi-Pin		J4	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, λSS: 0.24, σSS: 0.06, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.103	1.25T-1-WT-4A	Connection	Multi-Pin		J2	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λ G: 0.16, σ G: 0.04, π Q: 3, π E: 1, π T: 1, λ SS: 0.48, σ SS: 0.12, Temperature: 40
1.104	60076-5007-1001	Connection	Multi-Pin		J5	50*0.5mm- Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λ G: 2, σ G: 0.5, π Q: 3, π E: 1, π T: 1, λ SS: 6, σ SS: 1.5, Temperature: 40
1.105	C1402-1009-211	Connection	Multi-Pin		J2	193-F093K230-B06	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λ G: 0.4, σ G: 0.1, π Q: 3, π E: 1, π T: 1, λ SS: 1.2, σ SS: 0.3, Temperature: 40
1.106	1.25T-1-LT-10A	Connection	Multi-Pin		J9	10*1.25mm	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λ G: 0.4, σ G: 0.1, π Q: 3, π E: 1, π T: 1, λ SS: 1.2, σ SS: 0.3, Temperature: 40
1.107	1.25T-1-LT-8A	Connection	Multi-Pin		J6	8*1.25mm	1	0.001278	0.001278	Quality Level: Level I, Number of Pins: 8		Level I	λ G: 0.32, σ G: 0.08, π Q: 3, π E: 1, π T: 1, λ SS: 0.96, σ SS: 0.24, Temperature: 40
1.108	1.25T-1-LT-6A	Connection	Multi-Pin		J5	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λ G: 0.24, σ G: 0.06, π Q: 3, π E: 1, π T: 1, λ SS: 0.72, σ SS: 0.18, Temperature: 40
1.109	1.25T-1-LT-4A	Connection	Multi-Pin		J3	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λ G: 0.16, σ G: 0.04, π Q: 3, π E: 1, π T: 1, λ SS: 0.48, σ SS: 0.12, Temperature: 40
1.110	1.25T-1-LT-2A	Connection	Multi-Pin		J8	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, λ SS: 0.24, σ SS: 0.06, Temperature: 40
1.111	1.25T-1-LT-12A	Connection	Multi-Pin		J1	12*1.25mm	1	0.001916	0.001916	Quality Level: Level I, Number of Pins: 12		Level I	λ G: 0.48, σ G: 0.12, π Q: 3, π E: 1, π T: 1, λ SS: 1.44, σ SS: 0.36, Temperature: 40
1.112	60076-5007-1001	Connection	Multi-Pin		J11	50*0.5mm- Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λ G: 2, σ G: 0.5, π Q: 3, π E: 1, π T: 1, λ SS: 6, σ SS: 1.5, Temperature: 40
1.113	1.25T-1-WT-6A	Connection	Multi-Pin		J5	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λ G: 0.24, σ G: 0.06, π Q: 3, π E: 1, π T: 1, λ SS: 0.72, σ SS: 0.18, Temperature: 40

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Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.114	1.25T-1-LT-4A	Connection	Multi-Pin		J1	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λ G: 0.16, σ G: 0.04, π Q: 3, π E: 1, π T: 1, λ SS: 0.48, σ SS: 0.12, Temperature: 40
1.115	NRS3015T2R2 MNGH	Inductor	Coil	Chip/Ceramic Filter	L702,L703,L704	2.2uH-1.48A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λ G: 0.1, σ G: 0.07, π Q: 3, π E: 1, π T: 1, λ G: 0.1, σ G: 0.07, λ SS: 0.3, σ SS: 0.21, Temperature: 40
1.116	HXNR4030B-4R7M	Inductor	Coil	Chip/Ceramic Filter	L1,L2,L705	4.7uH-2A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λ G: 0.1, σ G: 0.07, π Q: 3, π E: 1, π T: 1, λ G: 0.1, σ G: 0.07, λ SS: 0.3, σ SS: 0.21, Temperature: 40
1.117	0.47mH-1.8A	Inductor	Coil	Chip/Ceramic Filter	L2	0.47mH-1.8A	1	0.000581	5.806969E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λ G: 0.1, σ G: 0.07, π Q: 3, π E: 1, π T: 1, λ G: 0.1, σ G: 0.07, λ SS: 0.3, σ SS: 0.21, Temperature: 40
1.118	P00805B-LV1	Semiconductor	Diode	General Purpose, Si	D1,D2,D9,D31,D85,D86	6.5V-4KV-SMB	6	0.004517	0.027101	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.001, Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.23566226, λ G: 4.6, σ G: 3.3, λ SS: 3.2521391, σ SS: 2.3330563, Temperature: 40
1.119	1SS355	Semiconductor	Diode	General Purpose, Si	D11,D69,D801	80V-0.1A-SOD323	3	0.000477	0.001432	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.31348618, λ G: 0.33, σ G: 0.23, λ SS: 0.31035132, σ SS: 0.21630546, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.120	SCD310H	Semiconductor	Diode	General Purpose, Si	D3,D4	100V-3A-2010/SMAF	2	0.001014	0.002028	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.61672421, λ G: 0.33, σ G: 0.23, λ SS: 0.61055697, σ SS: 0.42553971, Temperature: 40
1.121	BV-SMBJ20CAI	Semiconductor	Diode	General Purpose, Si	D5	20V-10/1000uS-800W-30V@8/20us-SMB	1	0.006982	0.006982	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.25837757, λ G: 4.6, σ G: 3.3, λ SS: 3.5656104, σ SS: 2.5579379, Temperature: 40
1.122	TUSD03FB	Semiconductor	Diode	TVS Surge Suppressor	D28,D29	3.3V-8/20uS-350W-Max19V-SOD323	2	0.002327	0.004655	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.001, Rated Forward Current: 8, Percentage: 0.0125	Rated Forward Current: 8, Percentage: 0.0125	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.23465534, λ G: 2.55, σ G: 0.81, λ SS: 1.7951133, σ SS: 0.57021246, Temperature: 40
1.123	RS1M-F1-0002	Semiconductor	Diode	General Purpose, Si	D3	1000V-1A-SMA	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.2, Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.41895155, λ G: 0.33, σ G: 0.23, λ SS: 0.41476203, σ SS: 0.28907657, Temperature: 40

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Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.124	1SS355	Semiconductor	Diode	General Purpose, Si	D4	80V-0.1A-SOD323	1	0.000600	5.995137E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.31348618, λ G: 0.33, σ G: 0.23, λ SS: 0.31035132, σ SS: 0.21630546, Temperature: 40
1.125	SCD310H	Semiconductor	Diode	General Purpose, Si	D7	100V-3A-2010/SMAF	1	0.001179	0.001179	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.3333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.3333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.61672421, λ G: 0.33, σ G: 0.23, λ SS: 0.61055697, σ SS: 0.42553971, Temperature: 40
1.126	BV-SMBJ58C2H	Semiconductor	Diode	TVS Surge Suppressor	D2,D5	58V-10us/700us-85MAX-SMB	2	0.002390	0.004781	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.2, Rated Forward Current: 21.37, Percentage: 0.93589143659	Rated Forward Current: 21.37, Percentage: 0.93589143659	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.24102391, λ G: 2.55, σ G: 0.81, λ SS: 1.8438329, σ SS: 0.5856881, Temperature: 40
1.127	SS510B-F1-0000HF	Semiconductor	Diode	General Purpose, Si	D1	100V-5A-SMB	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 5, Percentage: 20, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 5, Percentage: 20, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.41895155, λ G: 0.33, σ G: 0.23, λ SS: 0.41476203, σ SS: 0.28907657, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.128	MB6F	Semiconductor	Diode	General Purpose, Ge	Z1,Z2	Glass Passivated Single-Phase Bridge Rectifier	2	0.074260	0.148520	Type: General Purpose, Ge, Quality Level: Level I, Operating Current: 0.4, Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Level I	π Q: 3, π E: 1, π T: 1, π S: 2.3869109, λ G: 6.2, σ G: 4.4, λ SS: 44.396542, σ SS: 31.507223, Temperature: 40
1.129	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.130	S8550	Semiconductor	Transistor	Field Effect, Silicon	Q3	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.131	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q4,Q7,Q14	40V-0.5A-SOT23	3	0.003921	0.011762	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.132	YJS2305A-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q8	20V-5.4A-SOT23-6L	1	0.007293	0.007293	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.3, Rated Power: 1.2, Percentage: 25	Rated Power: 1.2, Percentage: 25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.54881164, λ G: 3.44, σ G: 0.75, λ SS: 5.6637361, σ SS: 1.2348262, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.133	BSS138W-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q3	50V-0.34A-SOT323	1	0.004697	0.004697	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.15, Percentage: 6.6666666667	Rated Power: 0.15, Percentage: 6.6666666667	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.35345468, λ G: 3.44, σ G: 0.75, λ SS: 3.6476523, σ SS: 0.79527303, Temperature: 40
1.134	S8550-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.135	NCE60P04SN	Semiconductor	Transistor	Field Effect, Silicon	Q3	60V-4A-SOT23-6L	1	0.008907	0.008907	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.5, Rated Power: 1.5, Percentage: 33.3333333333	Rated Power: 1.5, Percentage: 33.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.67032005, λ G: 3.44, σ G: 0.75, λ SS: 6.9177029, σ SS: 1.5082201, Temperature: 40
1.136	Light board	Optical Device	Other	Single LED / LCD Segment	LD1,LD2,LD3,L4	Night Mode	4	0.001110	0.004440	Type: Single LED / LCD Segment, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 0.25, σ G: 0.18, λ SS: 0.75, σ SS: 0.54, Temperature: 40
1.137	[HFD42/4.5-3SR(825)]-DCSV-HF	Relay	General Purpose		K1	HFD42/4.5-3SR(825)	1	0.009483	0.009483	Quality Level: Level I, Contact Current: 1, Rated Current: 4, Percentage: 25	Contact Current: 1, Rated Current: 4, Percentage: 25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.72252735, λ G: 1.9, σ G: 1.9, λ SS: 4.1184059, σ SS: 4.1184059, Temperature: 40
1.138	TSTP6650K-180BH-8.4	Switch	Pushbutton		SW1	DC12V-50mA-4PIN-6*6*5mm	1	0.007842	0.007842	Quality Level: Level I, Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.67705687, λ G: 1.99, σ G: 1.4, λ SS: 4.0420295, σ SS: 2.8436389, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.139	K-F-24.000M-9-2020-4085-80	Miscellaneous	Crystal Oscillator		X803	24M±20ppm-(-40°C-85°C)-3.2*2.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 3.2, σ G: 1.5, λ SS: 9.6, σ SS: 4.5, Temperature: 40
1.140	SF32WK32768D31T002	Miscellaneous	Crystal Oscillator		X802	32.768K±20ppm-(-40°C-85°C)-3.2*1.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 3.2, σ G: 1.5, λ SS: 9.6, σ SS: 4.5, Temperature: 40
1.141	YT30028FB.IR-JZ-DK-	Rotating Device				Lens-Electric Zoom	2	1.000000	2.000000				
1.142	IR0109	Rotating Device				Day and Night Mode	1	1.000000	1.000000				
1.143	ETA5052S2G	Integrated Circuit	Analog		U2	[ETA5052S2G/ETA5052V0S2G]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.144	LP3981AB5F]-SOT23-5	Integrated Circuit	Analog		U3,U7	[LP3981AB5F]-SOT23-5	2	0.026871	0.053741	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.145	[TP10-2-VR]-VSSOP8	Integrated Circuit	Analog		U4	[TP10-2-VR]-VSSOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.146	[TPM16050-S6TR/TMI8113]-SOT23-6	Integrated Circuit	Analog		U5	[TPM16050-S6TR/TMI8113]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.147	TP8485E-SR	Integrated Circuit	Analog		U5	[TP8485E-SR/SIT3088EESA]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.148	DIO54335TST6]-SOT23-6-dioo	Integrated Circuit	Analog		U709	[DIO54335TST6]-SOT23-6	1	0.028473	0.028473	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40
1.149	ETA3485S2F]-SOT23-5-ETA	Integrated Circuit	Analog		U702,U703,U706	[ETA3485S2F]-SOT23-5	3	0.022631	0.067893	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40
1.150	ETA5053V180DF1E]-1.8V-DFN1*1-ETA	Integrated Circuit	Analog		U710	[ETA5053V180DF1E]-1.8V-DFN1*1	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.151	LP3981AB5F]-SOT23-5	Integrated Circuit	Analog		U707	[LP3981AB5F]-SOT23-5	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.152	TMI5302]-TSOT23-6	Integrated Circuit	Analog		U18,U38	[TMI5302]-TSOT23-6	2	0.030756	0.061513	Quality Level: Level I, Number of Transistors: 768		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.1386501, σ G: 4.3400256, λ SS: 18.41595, σ SS: 13.020077, Temperature: 40
1.153	MS41908M	Integrated Circuit	Analog					#DIV/0!	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.154	[SD4952CTR]-SO16	Integrated Circuit	Analog		U1	[SD4952CTR]-SO16	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40

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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.155	IMX307LQR-C	Integrated Circuit	Digital		U1	[IMX307LQR-C]-2.0MP-1/2.8-LGA110	1	0.031007	0.031007	Quality Level: Level I, Technology: CMOS, Number of Gates: 5000000		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.9374105, σ G: 2.5470825, λ SS: 20.812232, σ SS: 7.6412476, Temperature: 40
1.156	W25N01GVZEIG	Integrated Circuit	ROM, PROM, EPROM		U1101	[W25N01GVZEIG]-WSON8 (8x6mm)-3V3	1	0.249449	0.249449	Quality Level: Level I, Technology: CMOS, Number of Bits: 1, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 42.756504, σ G: 30.228848, λ SS: 128.26951, σ SS: 90.686545, Temperature: 40
1.157	NT5CB256M16ER-FL	Integrated Circuit	RAM		U1001	[NT5CB256M16ER-FL]-FBGA96	1	0.115747	0.115747	Quality Level: Level I, Technology: Dynamic NMOS/CMOS, Number of Bits: 4, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 20.379802, σ G: 13.601017, λ SS: 61.139406, σ SS: 40.80305, Temperature: 40
1.158	C5-36GW	Integrated Circuit	Microprocessor		U801	[C5-36GW]-BGA307	1	0.154317	0.154317	Quality Level: Level I, Technology: CMOS, Bus Width in Bits: 32		Level I	π Q: 3, π E: 1, π T: 1, λ G: 24.604507, σ G: 20.224777, λ SS: 73.81352, σ SS: 60.674332, Temperature: 40
1.159	S16403	Inductor	Transformer	Power (> 1W)	T1	[S16403]100M POE-1 channel	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λ G: 5, σ G: 2, π Q: 3, π E: 1, π T: 1, λ G: 5, σ G: 2, λ SS: 15, σ SS: 6, Temperature: 40
1.160	GEP13	Inductor	Transformer	Power (> 1W)	T1	GEP13-127 μ H	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λ G: 5, σ G: 2, π Q: 3, π E: 1, π T: 1, λ G: 5, σ G: 2, λ SS: 15, σ SS: 6, Temperature: 40
1.161	[ML414H-IV01E]-3V	Miscellaneous	Battery		BAT801	Rechargeable, 1mAh	1	0.104891	0.104891	Battery Type: Lithium, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 25, σ G: 7.5, λ SS: 75, σ SS: 22.5
1.162	Fuse-4A	Miscellaneous	Fuse		FU1	[6125SB4A250V-G]-4A-250V 6125	1	0.001964	0.001964	Fuse Type: $\leq$ 30A, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 0.4, σ G: 0.19, λ SS: 1.2, σ SS: 0.57

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